

Compositions

11 trait levels

5 traits $\times$ high/low

4 heterogeneity

$\sigma = .05/.15/.25$

9 surface cells

$\mu_O \times \mu_A$

1 human-calibrated

IPIP-NEO-120 norms

$\theta_i \in [0, 1]^5$

O C E A N

validated

induction gate $\rho = .94$

Topics

gun control

contested

immigration

contested

filler topic

neutral control

6 estimation

World Bank truth

2 hidden profile

unshared information

Platform

$N = 100$

scale ablation 200

$T = 30$ rounds

activation $\rho = 0.4$

scale-free graph

follow / unfollow

interest + hot feed

size 10

bounded memory

$k = 10$

Models

Llama-3.1

8B primary

Qwen2.5

14B replication

vLLM

local serving

1 GPU, 24 GB

387 runs

temp 0.7

0.3 for probes

Outcomes

opinion variance

extremity

cross-cutting

echo closure

collective accuracy

hidden-profile rate

diversity term

Perturbations

2 topologies

small-world, random

activation $\pm 50\%$

memory $\times 2$

feed $\times 1.5$

temperature 1.0

model swap

sign stability

Replication

8 seeds

per condition

3 internal

identical compositions

2nd model family

E1, E2, E3 repeated

code + data

public, archived

387 runs

≈ 21 GPU-hours

Seven dimensions, one factorial programme: composition is varied, everything else is held fixed or perturbed deliberately